Volume 7 – Strategy Engine Design Document

AI Quant Trading Platform

Version: 1.0

Document Type: Strategy Engine Design

Classification: Core Trading Engine

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1. Purpose

The Strategy Engine is responsible for converting market intelligence into executable trading decisions.

While the AI Decision Engine evaluates probability and market conditions, the Strategy Engine determines **how** a trade should be executed according to predefined rules.

The Strategy Engine must be deterministic, configurable, explainable, testable, and fully compatible with paper trading and backtesting.

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2. Objectives

The Strategy Engine shall:

- Support multiple trading strategies simultaneously
- Allow users to enable or disable individual strategies
- Validate all trades through the Risk Engine
- Prevent conflicting strategy execution
- Adapt to different market regimes
- Generate structured trade plans before execution

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3. Strategy Execution Pipeline

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Market Scanner
      │
      ▼
AI Decision Engine
      │
      ▼
Market Regime Detection
      │
```

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      ▼
Eligible Strategy Selection
      │
      ▼
Entry Validation
      │
      ▼
Risk Validation
      │
      ▼
Position Sizing
      │
      ▼
Order Execution
      │
      ▼
Trade Monitoring
      │
      ▼
Exit Management
```

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4. Strategy Categories

Trend Following

Purpose

Capture medium and long-term directional trends.

Suitable Markets

- Strong Bull Markets
- Strong Bear Markets

Not Suitable

- Sideways Markets

Indicators

- EMA
- ADX
- SuperTrend
- VWAP

Breakout Strategy

Purpose

Trade price breaking important resistance or support.

Confirmation

- Volume Spike
- Candle Close
- Higher Timeframe Trend

Filters

- Fake Breakout Detection
- Volatility Check

Pullback Strategy

Purpose

Join an existing trend after temporary retracement.

Requirements

- Trend Confirmation
- Pullback Completion
- Momentum Recovery

Momentum Strategy

Purpose

Trade assets showing exceptional strength.

Inputs

- Volume
- RSI
- MACD
- ATR
- Price Acceleration

Swing Strategy

Holding Period

Several hours to several days.

Requirements

- Higher Timeframe Trend
- Strong Support
- Strong Resistance

Scalping Strategy

Holding Time

Seconds to Minutes

Requirements

- High Liquidity
- Tight Spread
- Low Slippage

Risk

Highest

Grid Trading

Purpose

Generate profits inside ranging markets.

Requirements

- Sideways Market
- Stable Volatility

Automatically places multiple buy/sell levels.

Dollar Cost Averaging (DCA)

Purpose

Gradually build positions.

Rules

- Maximum Number of Entries
- Minimum Distance Between Entries
- Maximum Exposure

Mean Reversion

Purpose

Trade reversion toward statistical averages.

Indicators

- Bollinger Bands
- RSI
- ATR

News Reaction Strategy

Purpose

React to significant market news.

Sources

- Exchange Announcements
- Token Listings
- ETF News
- Regulatory Updates

Smart Money Concepts (SMC)

Concepts

- Liquidity Sweep
- Order Blocks
- Fair Value Gap (FVG)
- Break of Structure (BoS)
- Change of Character (ChoCH)
- Premium / Discount Zones

ICT Concepts

Support

- Optimal Trade Entry
- Order Blocks
- Liquidity Pools
- Market Structure
- Session Timing

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5. Strategy Selection

The AI Decision Engine determines market conditions.

The Strategy Engine selects only strategies compatible with:

- Market Regime
- User Risk Profile
- Timeframe
- Portfolio Exposure
- Active Positions

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6. Entry Validation

Every strategy validates:

Trend

Momentum

Volume

Liquidity

Spread

Market Structure

Risk

Higher Timeframe Confirmation

News Filter

Exchange Availability

All conditions must pass before execution.

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7. Exit Management

Supported Exit Methods

Fixed Take Profit

Multiple Take Profit Levels

Trailing Stop

Time-Based Exit

Indicator Exit

Manual Exit

Emergency Exit

AI Recommendation Exit

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8. Stop Loss Management

Supported Methods

Fixed Percentage

ATR-Based

Swing High/Low

Support & Resistance

Order Block

Volatility Adaptive

Trailing Stop

Break-even Stop

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9. Position Sizing

Supported Models

Fixed Size

Fixed Risk %

Kelly Criterion (Experimental)

Volatility-Based

ATR-Based

Portfolio Percentage

Maximum Exposure

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10. Trade Filters

Reject trades if:

- Volume is insufficient
- Spread is excessive
- Volatility exceeds configured limits
- Higher timeframe conflicts
- News blackout period
- Daily loss limit reached
- Exchange unavailable
- Risk Engine rejects trade

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11. Trade Lifecycle

Candidate Generated

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Strategy Selected

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Entry Approved

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Risk Approved

↓

Position Opened

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Trade Monitored

↓

Exit Triggered

↓

Performance Stored

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Learning Dataset Updated

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12. Strategy Performance Tracking

Each strategy records:

- Total Trades
- Winning Trades
- Losing Trades
- Win Rate
- Average Profit
- Average Loss
- Profit Factor
- Drawdown
- Sharpe Ratio
- Average Holding Time

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13. AI Compatibility

The Strategy Engine consumes AI recommendations but remains rule-based.

AI provides:

- Confidence Score
- Risk Estimate

- Market Regime
- Opportunity Ranking

Strategy logic determines whether execution is allowed.

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14. Paper Trading Compatibility

Every strategy must support:

- Simulation Mode
- Historical Replay
- Backtesting
- Forward Testing

before live deployment.

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15. Configuration

Users may configure:

- Enabled Strategies
- Timeframes
- Risk Profile
- Maximum Trades
- Trading Sessions
- Preferred Assets
- Confidence Threshold
- Automation Level

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16. Safety Rules

Strategies shall never:

- Override the Risk Engine
- Ignore exchange restrictions
- Submit duplicate orders
- Trade during emergency stop
- Exceed exposure limits

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17. Future Strategies

Future versions may include:

- Statistical Arbitrage
- Pair Trading
- Cross-Exchange Arbitrage
- Market Making
- Options Strategies
- Volatility Strategies
- AI-Generated Strategies
- Reinforcement Learning Policies

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18. Design Principles

- Strategies are modular and independently deployable.
- Every strategy must expose explainable entry and exit criteria.
- All strategy decisions are logged and auditable.
- Strategy execution must be deterministic for identical inputs.
- New strategies can be added through a plugin architecture without modifying the core engine.